



NORTHERN UNIVERSITY

OF BUSINESS & TECHNOLOGY KHULNA

Department of Computer Science and Engineering

Project documentation

Course Title: Mobile Computing LAB

Course Code: CSE 4204

Project name: Medical Report Analyzer

Submitted To:

Md. Riaz Mahmud

Assistant Professor, Department of Computer Science and Engineering

Northern University of Business and Technology Khulna

Submitted By:

Team : 02

Section: 8D

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Team Members & Task Distribution

Student ID	Name	Role	Task Distribution
11220320899	Farjana Easmin Minka	AI Integration Lead & Database Manager	AI Integration
11220320959	Hassan A.K Azad Tushin	Team Leader	Documentation, Database Management, Testing
11220320982	Juena Tabassum	Frontend Developer	Frontend Development: React.js
11220321050	Faika Newaz Silmi	Backend Developer	Backend Development: Node.js, Express.js

Project Information

Project Overview

The Medical Report Analyzer is an AI-powered web application designed to bridge the gap between complex medical documentation and patient comprehension. By leveraging large language models, the system translates clinical terminology, lab values, diagnoses, and prescriptions into plain, readable language that any patient can understand without a medical background.

Objectives

- Provide patients with instant, plain-language summaries of their medical reports.
- Extract and highlight key findings such as diagnoses, lab values, and abnormal results.
- Offer an interactive chatbot for follow-up questions grounded in the uploaded report.
- Securely store reports, summaries, and chat histories per authenticated user account.

Support both text-pasted and image/scanned report uploads via OpenAI Vision

Core Features of the System

1. Medical Report Upload & Input

Users can upload medical reports in multiple formats, including text, PDF, and scanned images. The system processes and prepares these inputs for analysis.

2. AI-Powered Report Analysis

The system leverages AI to interpret medical reports and provide:

- Plain-language summaries for easier understanding
- Extraction of key findings such as diagnoses, lab values, and medications
- Identification and highlighting of abnormal or out-of-range results

3. Interactive Medical Chatbot

Users can engage in a context-aware chatbot that answers follow-up questions based on their specific medical report, enabling deeper understanding and clarification.

4. User Authentication & Account Management

Secure authentication is implemented using JWT, supporting:

- User registration
- Login and logout functionality

- Session management and protection of user data

5. Report History Management

Users can access, review, and manage all previously uploaded and analyzed medical reports within their account dashboard.

6. Data Storage & Security

All user data, including reports, summaries, and chat history, is securely stored using Supabase with strict access control policies to ensure privacy and data protection.

UI/Wireframe Screenshots

Fig: Landing Page

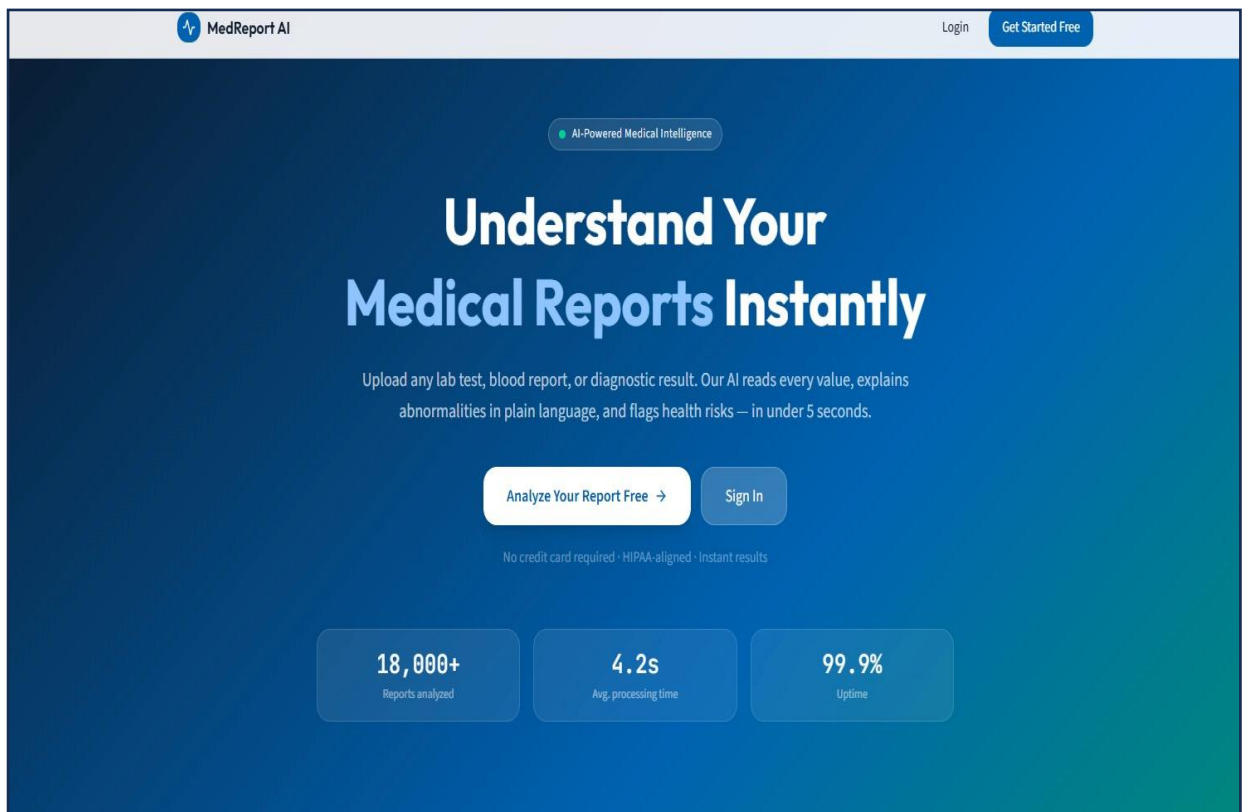


Fig: Registration page



Create Account

Start analyzing your medical reports

FULL NAME

Alex Rivera

EMAIL ADDRESS

your@email.com

PASSWORD

Min. 8 characters

AGE (optional)

34

EXISTING CONDITIONS (optional)

Diabetes

Hypertension

Heart Disease

Thyroid Disorder

Kidney Disease

Asthma

Create Account

Already have an account? [Sign in](#)

Fig: Login page



MedReport AI

Sign in to your account

EMAIL ADDRESS

alex@example.com

PASSWORD

.....

[Forgot password?](#)

Sign In

No account? [Create one](#)

Fig: Dashboard

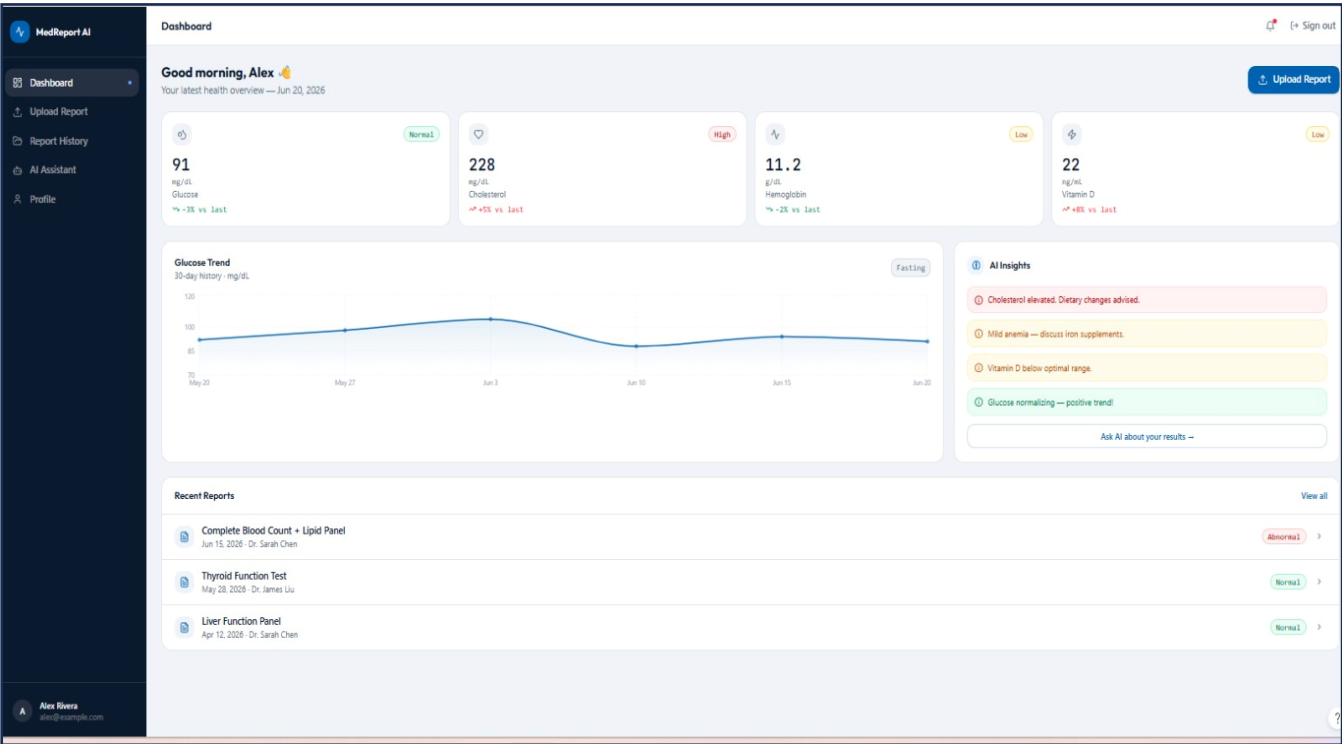


Fig: Upload report Page

Upload Medical Report

Supports PDF, JPG, PNG — any lab report format

Drop your report here

or click to browse files

PDF · JPG · PNG · up to 20 MB

End-to-end encrypted

Results in ~5 seconds

Any lab format

Analyze Demo Report →

No file selected? Click to analyze a sample CBC + Lipid Panel report

Fig: Result page

Complete Blood Count + Lipid Panel

Abnormal

Export

All Reports

Patient: Alex Rivera · Jun 15, 2026 · Dr. Sarah Chen, MD · Lab ID: LAB-2026-001

AI Health Summary

Plain language

Your blood test results show a few areas that need attention. You have **mild anemia** (hemoglobin 11.2 g/dL), which may explain fatigue. Your **cholesterol and LDL are elevated**, increasing long-term cardiovascular risk. Your **fasting glucose is in the pre-diabetic range** — worth addressing with diet changes. On the positive side, your kidney, liver, and thyroid function are all within normal limits.

5

4

3

12

Normal

Abnormal

Low

Total markers

Detected Values

12 markers

MARKER	VALUE	RANGE	STATUS
Hemoglobin	11.2 g/dL	13.5–17.5	Low
White Blood Cells	7.4 K/ μ L	4.5–11.0	Normal
Platelets	420 K/ μ L	150–400	High
Glucose (Fasting)	108 mg/dL	70–99	High
Creatinine	0.9 mg/dL	0.7–1.3	Normal
ALT	38 U/L	7–56	Normal
Cholesterol (Total)	228 mg/dL	<200	High
HDL Cholesterol	62 mg/dL	>40	Normal
LDL Cholesterol	142 mg/dL	<100	High
Triglycerides	156 mg/dL	<150	High
TSH	2.1 mIU/L	0.4–4.0	Normal

Risk Warnings

Elevated Cholesterol

Total cholesterol and LDL above recommended levels. Increase cardiovascular risk over time.

Pre-diabetic Glucose Range

Fasting glucose of 108 mg/dL is just above normal. Monitor with dietary changes.

Mild Anemia

Hemoglobin below normal range. May cause fatigue. Likely iron-related.

Vitamin D Deficiency

Below optimal range. Important for bone health, immunity, and mood.

Recommendations

Schedule follow-up with your doctor within 4 weeks

Reduce saturated fat and refined carbohydrate intake

Aim for 30 min of aerobic exercise 5+ per week

Eat iron-rich foods: lentils, spinach, lean red meat

Take Vitamin D3 supplement (1,000–2,000 IU/day)

Fig: Report History

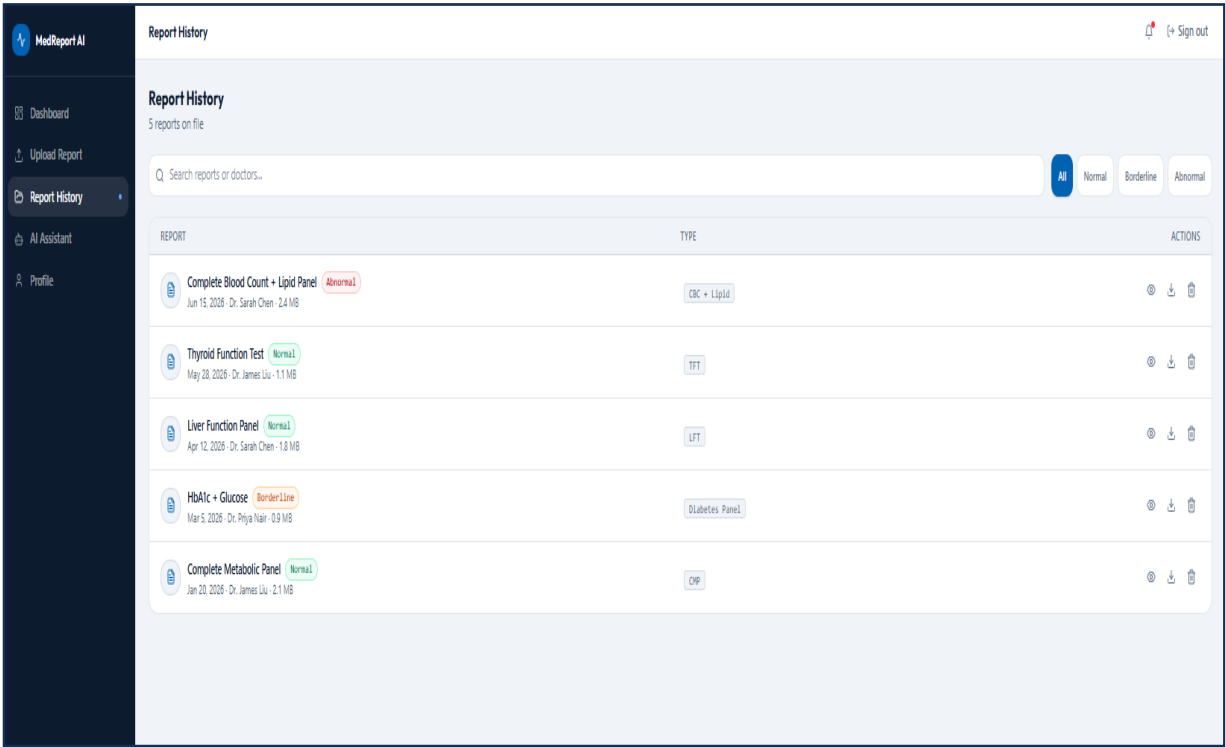


Fig: AI Assistant

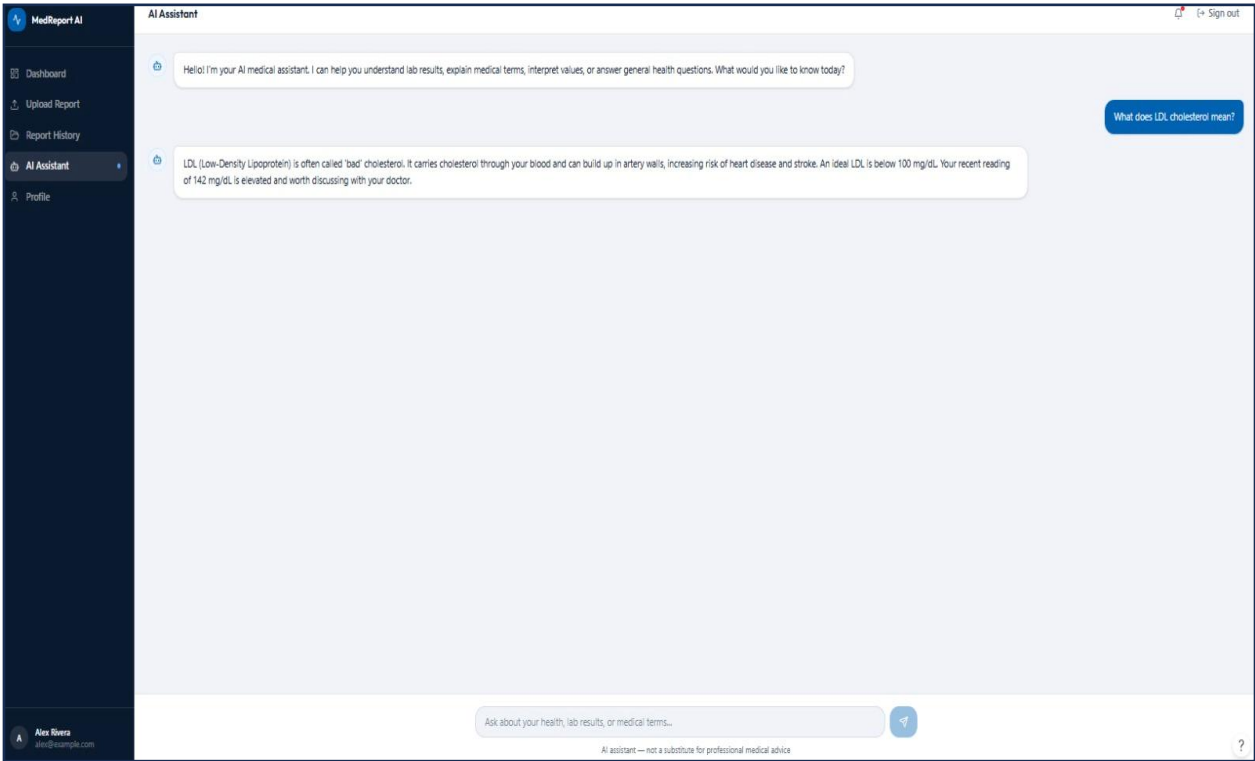


Fig: User Profile

MedReport AI

Dashboard

Upload Report

Report History

AI Assistant

Profile

Alex Riveraalex@example.com

Profile

Manage your account and health settings

Edit

Alex Riveraalex@example.comActive

FULL NAMEAlex Rivera

EMAILalex@example.com

PHONE+1 (555) 234-5678

AGE34

LOCATIONSan Francisco, CA

Medical Summary

5Total Reports

2Abnormal Findings

Jun 15Last Upload

4Monitored

Pre-diabetic monitoring

Vitamin D insufficiency

Cholesterol management

Iron deficiency anemia

Settings & Privacy

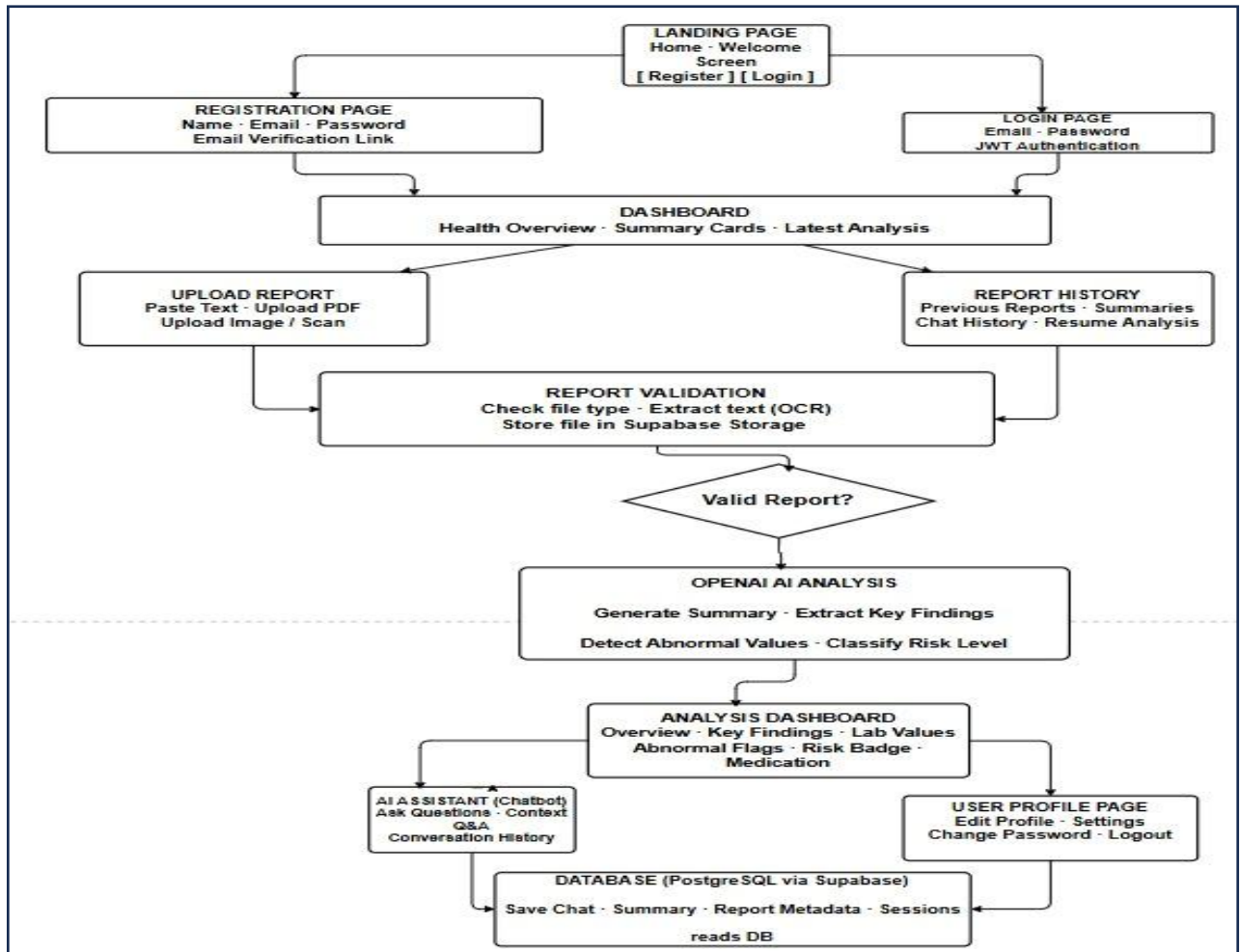
Email notifications for new analysis

Share de-identified data for research

Two-factor authentication

Change password

User Workflow Diagram



Medical Report Analyzer AI — Project Roadmap

WEEK 6 — Backend Development

1. Authentication System

Build secure user access:

Features:

- User Registration
- Login
- JWT-based authentication
- Password hashing

Tech Stack:

- Node.js + Express
- bcrypt (password hashing)
- JSON Web Token (JWT)

API Endpoints:

- POST /register
- POST /login
- GET /profile

2. Database Integration

Database :

PostgreSQL

Collections/Tables:

- Users
- MedicalReports
- AIResults

Key Fields Example (MedicalReports):

- userId
- reportText / fileURL
- uploadedAt
- extractedData (AI output)

3. REST APIs

Core APIs for project:

- POST /upload-report → upload medical report (PDF/text/image)
- POST /analyze-report → send data to AI model
- GET /reports → fetch user reports
- GET /report/:id → get single analysis

Deliverables:

- Backend source code
- API documentation (Postman collection recommended)

- Database schema

WEEK 7 — Frontend Development

1. User Interface (React)

Pages to build:

- Login / Register page
- Dashboard (main panel)
- Upload Medical Report page
- Report Analysis View page
- AI Chat/Assistant screen

2. API Integration

Connect frontend with backend:

Handle:

- Loading states (spinner during AI processing)
- Error handling (invalid file, API failure)
- Success messages (analysis completed)
- Form validation

3. Responsive Design

Use:

- React + Tailwind CSS

Must support:

- Mobile 
- Tablet 
- Desktop 



Deliverables:

- Full frontend code
- Connected backend APIs

WEEK 8 — AI Integration (Core Phase)

1. AI Workflow

User Upload Report



Backend Processes Data



AI Model/API (OpenAI or Gemini)



Extract & Analyze Report



Return Structured Result



Frontend Displays Output

2. AI Features for Medical Report Analyzer

⚠ Important: This system should NOT diagnose diseases, only:

- Extract information
- Summarize reports
- Highlight abnormal values (if instructed)
- Provide simplified explanations

3. AI Functional Modules



1. Report Summarizer

- Converts complex report into simple language



2. Medical Term Explainer

- Explains medical terms in easy language



3. Structured Data Extractor

Example output:

```
{
  "hemoglobin": "12.5 g/dL",
  "glucose": "110 mg/dL",
  "notes": "Slightly above normal range"
}
```



4. AI Chat Assistant

- User asks: “What does this report mean?”
- AI responds in simple explanation

4. Prompt Engineering

Bad Prompt:

Explain this report

Good Prompt:

Summarize this medical report in simple terms for a non-medical university student.

Highlight only key values and explain them without diagnosing any disease.

5. AI APIs

- OpenAI API
- Google Gemini API

Deliverables:

- AI workflow diagram
- AI integration code
- Prompt documentation

WEEK 9 — Feature Completion

Final System Integration Complete:

- Authentication fully working
- Report upload + storage
- AI analysis working smoothly
- Dashboard UI polished
- Navigation fixed

System Checklist

- ✓ Login/Register works
- ✓ File upload works
- ✓ AI returns response correctly
- ✓ Database stores reports
- ✓ UI responsive
- ✓ No broken routes

Optimization Tasks

- Reduce API response time
- Improve prompt efficiency

Deliverables:

- Near-final project
- Feature checklist report

WEEK 10 — Testing & Debugging

1. Functional Testing

- Login/register validation
- Upload and analysis flow

2. API Testing

Use:

- Postman

Test:

- All endpoints
- Error handling
- Response structure

3. UI Testing

Check:

- Mobile responsiveness
- Button interactions
- Form validation

4. AI Testing Verify:

- Correct summarization
- No hallucinated medical claims
- Proper error handling for empty/invalid input

Final Deliverables:

- Fully working project
- Documentation (API + AI flow + DB schema)
- Demo-ready system

Technology Stack

Frontend	React.js
Backend	Node.js, Express.js
Database	PostgreSQL
AI Services	OpenAI